

Waldspurger on Average (Alan+Jeff)

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Abstract

Fix an integer $k \geq 3$, let f be a half-integral weight holomorphic cusp form of weight $\lambda = k + \frac{1}{2}$ on $\Gamma_0(4)$ with Fourier coefficients $a_f(n)$, and let F be its weight $2k$ Shimura correspondent on $\Gamma_0(1)$. Also let D be a fundamental discriminant satisfying $(-1)^k D > 0$. Then the Kohnen-Zagier formula says

$$a_f(|D|)^2 = \frac{\Gamma(k)}{\pi^k} |D|^{k-\frac{1}{2}} L\left(\frac{1}{2}, F \times \chi_D\right).$$

We will prove this formula, in an average sense, using various Petersson trace formulas related to integral and half-integral weight holomorphic forms.

1 Heuristics

We will prove a formula roughly of shape

$$\sum_f \sum_d \frac{a_F(|D|)^2}{D^w} \approx \sum_F \sum_D \frac{L\left(\frac{1}{2}, F \times \chi_D\right)}{D^w},$$

and then to isolate individual terms, or short averages, via Perron-type integrals. We should expect a formula like this to be true for the following reason (and for heuristic simplicity we ignore constants and the explicit dependencies on sums): there is a rough multiple Dirichlet series of shape

$$Z(w, s; F) = \sum_{m,d} \frac{a_F(m) \chi_d(m)}{m^s d^w},$$

admitting the interchange formula

$$Z(w, s; F) = \sum_m \frac{a_F(m) L(w, \chi_m)}{m^s} = \sum_d \frac{L(s, F \times \chi_d)}{d^w}.$$

Now consider the two averages

$$\frac{\Gamma(2k-1)}{(4\pi)^{2k-1}} \sum_F a_F(m) \overline{a_F(n)} \quad \text{and} \quad \frac{\Gamma(\lambda-1)}{2(4\pi)^{\lambda-1}} \sum_f a_f(m) a_f(n).$$

these are the spectral sides of the integral and half-integral weight Petersson trace formulas. To the integral weight spectral sum, we set $n = 1$ and turn it into a Dirichlet series upon multiplying by $\frac{L^{(2)}(w, \chi_m)}{m^s}$ and summing over m . This gives

$$\sum_F \sum_m \frac{a_F(m) L^{(2)}(w, \chi_m)}{m^s}.$$

Upon applying the interchange for $Z(w, s; F)$, this double sum becomes

$$\sum_F \sum_d \frac{L^{(2)}(s, F \times \chi_d)}{d^w}.$$

To the half-integral weight spectral sum, we set $m = n = d$ and turn it into a Dirichlet series upon multiplying by $\frac{1}{d^w}$ and summing over d . We obtain

$$\sum_f \sum_d \frac{|a_f(d)|^2}{d^w}.$$

Letting $s = \frac{1}{2}$ in the integral weight sum (which is not formally allowed since the sum over m does not converge absolutely at this point), we expect these two latter double sums should be equal by the Kohnen-Zagier formula giving

$$\sum_f \sum_d \frac{|a_f(d)|^2}{d^w} = \sum_F \sum_d \frac{L\left(\frac{1}{2}, F \times \chi_d\right)}{d^w}.$$

Now that we expect this equality to be true, we seek a method of proof that avoids using the Kohnen-Zagier formula directly. The apparent avenue is to equate the corresponding geometric sums of the Petersson trace formulas in order to prove

$$\sum_f \sum_d \frac{|a_f(d)|^2}{d^w} = \sum_F \sum_m \frac{a_F(m) L(w, \chi_m)}{m^s},$$

and then apply the interchange for $Z(w, s; F)$ to the right-hand side to finish the argument.

2 Integral weight

Consider weight $2k$ holomorphic cusp forms on $\Gamma_0(1)$. We may choose a Hecke eigenbasis whose elements F are the Shimura correspondents of a basis for the Kohnen plus space $\Gamma_0(4)$ whose elements f are eigenfunctions for the square-free Hecke operators. For such an F , we write

$$F(z) = \sum_n a_F(n) n^{\frac{2k-1}{2}} e(nz),$$

for its Fourier series expansion. The integral weight Petersson trace formula says

$$\frac{\Gamma(2k-1)}{(4\pi)^{2k-1}} \sum_F a_F(m) \overline{a_F(n)} = \delta_{m,n} + (2\pi i)^{-2k} \sum_c \frac{K(m, n, c)}{c} J_{2k-1} \left(\frac{4\pi \sqrt{mn}}{c} \right).$$

Set $n = 1$, multiply by $\frac{L(w, \chi_m)}{m^s}$ where $L(w, \chi_m)$ is understood to be the corresponding L -function times a correction polynomial when m is not square-free, and summing over m . The spectral sum becomes

$$\frac{\Gamma(2k-1)}{(4\pi)^{2k-1}} \sum_F \sum_m \frac{a_F(m) L(w, \chi_m)}{m^s}.$$

As for the geometric sum, the diagonal term only contributes when $m = 1$ in which case $L(w, \chi_m) = \zeta(s)$. Thus we obtain

$$\zeta(w) + (2\pi i)^{-2k} \sum_m \sum_c \frac{L(w, \chi_m) K(m, 1, c) J_{2k-1} \left(\frac{4\pi\sqrt{m}}{c} \right)}{cm^s}.$$

We will study the off-diagonal contribution by expanding the Kloosterman sum and using the inverse Mellin transform representation of the J -Bessel function. Recall that the Kloosterman sum is

$$K(m, 1, c) = \sum'_{a \pmod{c}} e \left(\frac{am + \bar{a}}{c} \right).$$

We will also make use of the representation of the J -Bessel function as an inverse Mellin transform given by

$$J_{2k-1} \left(\frac{4\pi\sqrt{m}}{c} \right) = \frac{1}{4\pi i} \int_{(\gamma)} \frac{\Gamma(k + \frac{t}{2})}{\Gamma(k + \frac{1-t}{2})} \left(\frac{2\pi\sqrt{m}}{c} \right)^{-t} dt,$$

provided $0 > \gamma > 1 - k$. Applying these formulas and interchanging sums, the off-diagonal term may be expressed as

$$(2\pi i)^{-2k} \sum_c \frac{1}{c} \sum'_{a \pmod{c}} e \left(\frac{\bar{a}}{c} \right) \frac{1}{4\pi i} \int_{(\gamma)} \frac{\Gamma(k + \frac{t}{2})}{\Gamma(k + \frac{1-t}{2})} \left(\frac{2\pi}{c} \right)^{-t} Z^+ \left(w, s; \frac{a}{c} \right),$$

where

$$Z^\pm \left(w, s; \frac{a}{c} \right) = \sum_{\substack{n \\ \text{sgn}(n) = \pm}} \frac{e \left(\frac{an}{c} \right) L(w, \chi_n)}{|n|^{s + \frac{t}{2}}} dt.$$

This is an additively twisted double Dirichlet series. It is essentially the Mellin transform of a half-integral weight Eisenstein series and so possesses a natural functional equation. This series was first studied by Hoffstein and Goldfeld in 1985 (see **Todo: [see H-G in which $\delta = s$ and $\rho = w$]**). We will only discuss the relevant series and functional equations necessary. First write $m = nc + r$ with r taken modulo c so that

$$Z^\pm \left(w, s; \frac{a}{c} \right) = \sum_{r \pmod{c}} e \left(\frac{ar}{c} \right) Z_r^\pm(w, s; c),$$

where

$$Z_r^\pm(w, s; c) = \sum_{\substack{n \equiv r \pmod{c} \\ \text{sgn}(n) = \pm}} \frac{L(w, \chi_n)}{|n|^s}.$$

The aforementioned functional equation is more easily described for this series. To do so, set

$$\Phi_0^* \left(w, s, \frac{a}{c}; k \right) = c_k(w, s) \left[G_{+k}(w, s) Z^+ \left(s+1, w, \frac{a}{c} \right) + G_{-k}(w, s) Z^- \left(s+1, w, \frac{a}{c} \right) \right],$$

where we put

$$c_k(w, s) = i^{-\frac{k}{2}} \pi^{-s} 2^{\frac{k}{2}-2s-2w-1},$$

and $G_{\pm k}(w, s)$ satisfies

$$G_{\pm k}(w, s) = 2^{\frac{2w+1+k}{4}+s} \frac{\Gamma \left(\frac{2w+1+k}{4} + s \right)}{\Gamma \left(\frac{2w+1+k}{4} \right)} \left(1 + O_{w,k} \left(\frac{1}{|s|} \right) \right).$$

Further letting

$$\phi_r(w, s; k) = \frac{1}{c} \sum_{a \pmod{r}} e \left(-\frac{ar}{c} \right) \Phi_0^* \left(w, s, \frac{a}{c}; k \right),$$

orthogonality of characters then implies the relation

$$\phi_r(w, s; k) = c_k(w, s) \left[G_{+k}(w, s) Z_r^+(w, s; c) + G_{-k}(w, s) Z_r^-(w, s; c) \right],$$

This identity can be solved for $Z_r^\pm(w, s; c)$ as in **Todo: [H-G]** to obtain

$$\begin{aligned} Z_r^\pm(w, s; c) = \pm D_k(w, s)^{-1} & \left[\phi_r(w, s; k) G_{\mp(k+4)}(w, s) c_k(w, s)^{-1} \right. \\ & \left. - \phi_r(w, s, k+4) G_{\mp k}(w, s) c_{k+4}(w, s)^{-1} \right], \end{aligned}$$

where

$$D_k(w, s) = G_k(w, s) G_{-(k+4)}(w, s) - G_{k+4}(w, s) G_{-k}(w, s).$$

The functional equation for $Z_r^\pm$ arises from that of Φ_0^* . First set

$$\Phi_0^* \left(w, s, \frac{a}{c}; k \right) = \Phi_0 \left(\frac{2w+1+k}{4} + s, \frac{2w-k+1}{4}, \frac{a}{c}; k \right).$$

There will be three cases depending upon congruence conditions of c modulo 4:

1. Suppose $c \equiv 1, 3 \pmod{4}$ and d modulo c is chosen so that $-4ad \equiv 1 \pmod{c}$. Then

$$\Phi_0^* \left(w, s, \frac{a}{c}; k \right) = (2c)^{-2s-w-\frac{1}{2}} (-2i)^{\frac{k}{2}} \varepsilon_c^{-k} \left(\frac{d}{c} \right) \Phi_\infty^* \left(-w, -s-w-\frac{1}{2}, \frac{d}{c}; k \right),$$

where

$$\Phi_\infty^* \left(-w, -s-w-\frac{1}{2}, \frac{d}{c}; k \right) =$$

2. Suppose $c \equiv 0 \pmod{4}$ and d modulo c is chosen so that $-4ad \equiv 1 \pmod{c}$:
3. Suppose $c \equiv 2 \pmod{4}$ and d modulo c is chosen so that $-4ad \equiv 1 \pmod{c}$:

3 Half-integral weight

Recall $\lambda = k + \frac{1}{2}$ and suppose $(-1)^k m, (-1)^k n \equiv 0, 1 \pmod{4}$. The Petersson trace formula relative to a basis for the Kohnen plus space associated holomorphic cusp forms on $\Gamma_0(4)$ is given by

$$\frac{\Gamma(\lambda - 1)}{2(4\pi)^{\lambda-1}} \sum_f a_f(m) a_f(n) = \frac{2}{3} \left(\delta_{m,n} + 2\pi e^{-\frac{2\pi i \lambda}{4}} \sum_{c \geq 1} \frac{K_\lambda^+(m, n, c)}{c} J_{\lambda-1} \left(\frac{4\pi \sqrt{mn}}{c} \right) \right),$$

where

$$K_\lambda^+(m, n, c) = \sum_{\substack{d \pmod{c} \\ (c,d)=1}} \varepsilon_d^{2\lambda} \left(\frac{c}{d} \right) e^{\frac{2\pi i (md + n\bar{d})}{c}} \cdot \begin{cases} 0 & \text{if } 4 \mid c, \\ 2 & \text{if } 4 \mid c \text{ and } 8 \nmid c, \\ 1 & \text{if } 8 \mid c, \end{cases}$$

and ε_d and $\left(\frac{c}{d} \right)$ are as in the theta multiplier. **Todo:** [XXX]